Southern New Hampshire University

CS 499 Sample Exemplar Status Checkpoints for All Categories

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Checkpoint	Software Design and Engineering	Algorithms and Data Structures	Databases
Name of Artifact Used	**Artifact name:** Inventory Management Android Mobile App **Origin:** CS 360 Mobile Architect and Programming	**Artifact name:** Inventory Management Android Mobile App **Origin:** CS 360 Mobile Architect and Programming	**Artifact name:** Inventory Management Android Mobile App **Origin:** CS 360 Mobile Architect and Programming
Status of Initial Enhancement	Enhancements completed	Planned but not yet completed	Planned but not yet completed
Submission Status		Planned but not yet completed	Planned but not yet completed
Status of Final Enhancement		Planned but not yet completed	Planned but not yet completed
Uploaded to ePortfolio	Completed with polished narrative and confirmed navigation on GitHub Pages site	Planned but not yet completed	Planned but not yet completed
Status of Finalized ePortfolio	Ready for review in Module Seven	Planned but not yet completed	Planned but not yet completed

The artifact I selected is an Android inventory management application that I originally created in a previous mobile development course. The application allows users to create an account, log in, and manage inventory stored in a local SQLite database. Users can add and remove items, change quantities, and receive feedback when an item reaches zero.

I selected this artifact because it does a good job highlighting different areas of software development. The original application was functional, but there were glaring flaws and too much of the logic was contained in MainActivity. I improved the application with this enhancement by creating an InventoryItem model, separating the login and inventory functionality into different activities, upgrading input validation, and improving navigation and the user interface. I also began restructuring the inventory display using a RecyclerView in hopes of having a more maintainable app (GeeksforGeeks, 2025).

I believe this enhancement met the outcomes I originally selected. The process required me to evaluate an existing application and make decisions about how its design could be improved. I also used Android development techniques and object-oriented design principles to make the application more maintainable. I do not currently have any major changes to my planned course outcomes.

The biggest thing I learned from this enhancement was just how much of the code could be changed without actually showing any visual changes. Separating responsibilities into different classes made it easier to understand what each part of the application was responsible for. One challenge was making major structural changes without breaking code that already worked.

Below is a checklist of enhancements I've added to throughout the past week or so:

Created an InventoryItem model class

- Encapsulates ID, name, and quantity into a single object.
- Added constructor, getters and setters.

Refactored DatabaseHelper

- Changed getAllItems() from returning a raw Cursor to an ArrayList<InventoryItem>.
- DatabaseHelper now converts database rows into objects itself.
- Cursor creation, reading, and closing stays within the database layer.

Separated Login and Inventory responsibilities

- MainActivity now focuses on login/account creation.
- Created a separate InventoryActivity.
- Successful login uses an Intent to navigate to InventoryActivity.
- Moved inventory display and controls out of MainActivity.

Improved input validation

- Empty item information is rejected.
- try/catch prevents invalid quantities from crashing the app.
- Negative quantities are rejected.
- Invalid input gives the user feedback.

Improved navigation

- Login now navigates to a separate Inventory Activity instead of swapping layouts with setContentView().
- Added logout functionality.
- Logout clears the Activity stack so Back does not reopen the inventory screen.

Added a proper inventory toolbar/menu

- Added an Android Toolbar to the inventory screen.
- Added a three-dot overflow menu.
- Created a Logout button within said menu.

UI Enhancements

Inventory Add Item Flow

- Moved the "Add Item" button to the top of the activity_inventory.xml layout.
- Moved the name and quantity input fields below the button and set them to be hidden by default.
- Added a "Confirm Add" button inside the input section.

Dynamic Button Text

- Updated InventoryActivity.java to toggle the "Add Item" button text to "Cancel" when the inputs are shown, and back to "Add Item" when they are hidden.

Button Preservation

- Fixed an issue where long item names pushed the quantity field, -, +, and Remove buttons off the screen by using layout weights.

Logout Menu Logic

- Implemented the onOptionsItemSelected logic to handle logging out, which clears the activity stack and returns the user to the MainActivity (Login screen).

Scrolling Functionality

- Added scrolling functionality when the item list gets too large to fit the screen.

Improved Inventory List Architecture (This part was very difficult and was an idea I found online, below is a site that helped me learn how to implement a RecyclerView)

- Replaced the manually generated inventory rows with a RecyclerView.
- Created a custom InventoryAdapter to connect InventoryItem objects to the inventory display.
- Created a reusable XML layout for individual inventory items.
- Separated inventory display logic from InventoryActivity.

Reference

GeeksforGeeks. (2025, January 29). *RecyclerView in Android with example*. https://www.geeksforgeeks.org/android/android-recyclerview/